

Software Estimation in Software Project Management

Software estimation is one of the most basic and important activities in **Software Project Management (SPM)**. Before starting a project, the manager needs to estimate **how much work is required, how long it will take, how many people are needed, and how much it will cost**.

Simple Definition

Software estimation is the process of predicting the effort, time, resources, and cost required to develop a software project.

1. Why Do We Need Software Estimation?

Imagine a client says:

"Build me an online shopping website."

The project manager cannot simply start coding. They need to answer:

- How much software needs to be developed?
- How many developers are required?
- How long will development take?
- How much will it cost?
- What resources are required?
- What risks can affect the estimate?

Therefore, estimation is done **before and during the project**.

2. What Do We Estimate?

There are four major things students should remember:

Size → Effort → Time → Cost

```
Software Size
  ↓
  Effort
  ↓
Development Time
  ↓
  Cost
```

These are closely related.

A. Software Size

Size represents how large the software is.

Common measures include:

- **LOC** – Lines of Code
- **KLOC** – Thousand Lines of Code
- **Function Points (FP)**
- **Use Case Points (UCP)**

Example

If estimated software size is:

20 KLOC

it means approximately **20,000 lines of source code**.

Important: LOC is usually an estimate, not something known accurately before development.

3. Effort Estimation

Effort means the amount of human work required to complete the project.

It is commonly measured in:

- Person-hours
- Person-days
- Person-months

Example

Suppose:

5 developers × 4 months = 20 person-months

So estimated effort = **20 person-months**.

Important:

Effort ≠ Duration

If a project requires 20 person-months, it does **not automatically mean** that 20 people can finish it in one month.

Some tasks depend on each other, and adding people can also increase communication overhead.

4. Time / Schedule Estimation

This determines **how long the project will take**.

Example:

Estimated duration = 6 months

The project manager creates a schedule containing:

- Requirements
 - Design
 - Development
 - Testing
 - Deployment
-

5. Cost Estimation

Cost estimation determines the **total money required** for the project.

Costs can include:

- Developer salaries
- Testing
- Hardware
- Cloud services
- Software licenses
- Office/infrastructure
- Training
- Maintenance
- Contingency

Simple Example

Suppose:

Cost	Amount
------	--------

Cost	Amount
Development	₹10 lakh
Testing	₹2 lakh
Cloud/Hardware	₹1 lakh
Other expenses	₹1 lakh
Total	₹14 lakh

Estimated project cost = **₹14 lakh**.

6. Main Software Estimation Techniques

There are several ways to estimate software projects.

1. Expert Judgment

An experienced developer/project manager estimates the project based on previous experience.

Example:

"I have developed similar systems before. This project will probably require around 8 person-months."

Advantage:

Simple and quick.

Disadvantage:

Depends heavily on the expert's experience and can be subjective.

2. Estimation by Analogy

The current project is compared with a **similar completed project**.

Example:

Previous project:

College Management System → 10 person-months

New project is similar but slightly larger.

Estimated effort might be around:

12–14 person-months.

Advantage:

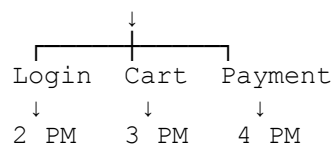
Useful when historical project data is available.

3. Decomposition Technique

The project is divided into smaller components, and each component is estimated separately.

Example:

Online Shopping System



Total:

2 + 3 + 4 = 9 person-months

This is called **bottom-up estimation**.

4. LOC-Based Estimation

Estimate the number of lines of code and then calculate effort.

For example:

Estimated size:

20 KLOC

If historical data tells us that a particular team requires a certain amount of effort for each KLOC, that information can be used to estimate total effort.

Problem:

It is difficult to accurately predict LOC before development.

5. Function Point (FP) Estimation

Function Point measures software size based on the **functionality provided to the user**, rather than lines of code.

It considers things such as:

- Inputs
- Outputs
- User inquiries
- Internal files
- External interfaces

Example

For a banking application:

- Login
- Deposit
- Withdrawal
- Balance inquiry
- Transaction statement

Each contributes to the functional size of the system.

Advantage:

It can be used before the programming language is selected.

6. COCOMO Model

COCOMO = Constructive Cost Model

It was developed by **Barry Boehm**.

COCOMO estimates:

- Effort
- Development time
- Project cost

based mainly on estimated software size.

The basic COCOMO effort equation is:

$$E=a(KLOC)^b$$

Where:

- **E** = Effort in person-months
- **KLOC** = Thousand Lines of Code
- **a, b** = model constants

For basic COCOMO, projects are commonly categorized as:

- Organic
 - Semi-detached
 - Embedded
-

7. Three-Point Estimation

Instead of making one estimate, three estimates are considered:

Optimistic (O)

Best-case estimate.

Most Likely (M)

Most realistic estimate.

Pessimistic (P)

Worst-case estimate.

A commonly used PERT-style expected estimate is:

$$E=6O+4M+P$$

Example

Suppose a task has:

- O = 4 days
- M = 6 days
- P = 10 days

Then:

$$E=64+4(6)+10 \quad E=638 \approx 6.33 \text{ days}$$

This technique considers **uncertainty and risk**.

8. What Affects Software Estimation?

Estimation is affected by many factors.

1. Project Size

Larger projects generally require more effort.

2. Complexity

A simple website is easier to estimate than a banking or AI system.

3. Team Experience

Experienced developers may complete tasks faster.

4. Technology

New technologies may introduce uncertainty.

5. Requirements

Changing or unclear requirements make estimation difficult.

6. Quality Requirements

Higher security, performance, and reliability requirements increase effort.

7. Tools and Infrastructure

Availability of development tools and infrastructure affects productivity.

8. Risk

Technical, organizational, and external risks can increase effort and cost.

9. Estimation vs Planning

Students often confuse these two.

Estimation

Answers:

"How much will the project require?"

Examples:

- 20 person-months
- 6 months
- ₹15 lakh

Planning

Answers:

"How will we complete the project?"

Planning includes:

- Assigning people
- Creating schedule
- Assigning tasks
- Setting milestones
- Managing resources

Simple Difference

Estimation = Predict

Planning = Organize

10. Estimation vs Scheduling

Estimation

Project requires **20 person-months** of effort.

Scheduling

Development will run from January to May, with 5 developers working on different tasks.

So:

Estimation tells us the expected effort/time/cost.

Scheduling decides when activities will happen and who will perform them.

11. Estimation Process

A simple software estimation process is:

```
Understand Requirements
      ↓
Estimate Software Size
      ↓
Estimate Effort
      ↓
Estimate Schedule
      ↓
Estimate Cost
      ↓
Review Risks & Assumptions
      ↓
Finalize Estimate
```

12. Important Point: Estimation Is Not Exact

Software estimation is a **prediction**, not a guarantee.

For example, saying:

"The project will definitely take exactly 5 months."

is usually unrealistic.

A better approach is:

"Based on current requirements and assumptions, the estimated duration is approximately 5 months."

As the project progresses and more information becomes available, estimates can be **revised**.

13. Example

Suppose a company wants to develop a **College Management System**.

The manager estimates:

- Software size = 30 KLOC
- Effort = 25 person-months
- Duration = 6 months
- Team = 5 people
- Development cost = ₹12 lakh
- Testing cost = ₹2 lakh
- Infrastructure = ₹1 lakh

Total Estimated Cost

$$\text{₹12L} + \text{₹2L} + \text{₹1L} = \text{₹15 lakh}$$

The manager can now prepare the project plan and budget.